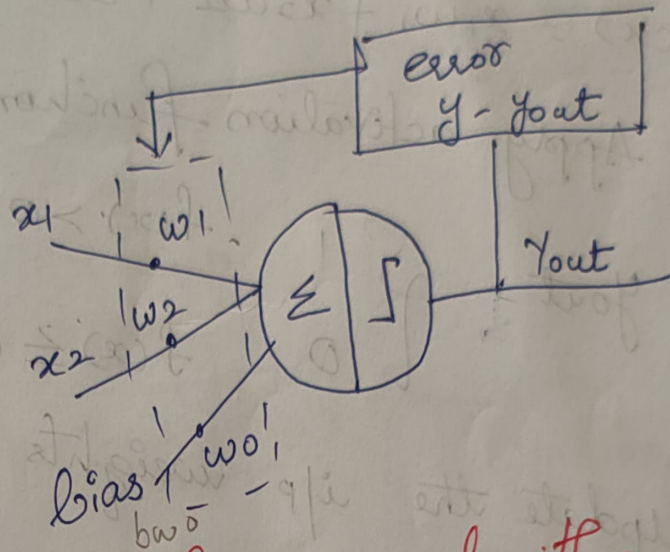


Construct a single layer perceptron for implementing OR gate using perceptron learning algorithm.

solo

data

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	1



Steps for perceptron learning algorithm.

i) identify no of inputs & o/p classes

OR - gate implementation

$x_1, x_2 \rightarrow$ i/p $y =$ o/p [2 classes]
 $b_1 = 1$ $\alpha = 0.4$ $\begin{bmatrix} 0, 1 \end{bmatrix}$

steps 2 Initialize mfw parameters.

i) Epochs ii) Bias input iii) learning rate

iv) i/p & bias weights update.

$\downarrow$
 $0 \rightarrow$ slow convergence

Perceptron learning algorithm data $1 \rightarrow$ fast
 rule aims to adjust weights based
 on error.

steps start the training process.

- i) Compute the net input given to the neuron

$$f(x) = x_1 w_1 + x_2 w_2 + \text{bias}(w_0) \rightarrow 0.5 = 1$$

- ii) Apply activation function (unit step, sigmoid)

$$y_{\text{out}} = \begin{cases} 1 & f(x) > 0 \\ 0 & f(x) \leq 0 \end{cases}$$

$$\rightarrow 0.2 \quad 0$$

- iii) Update the i/p weights & bias weights based on error

$$\text{Error} = y - y_{\text{out}}$$

- iv) The new updated weights are represented as

$$W_{\text{new}} = W_{\text{old}} + \alpha * \text{error} * x_i$$

$x_i \rightarrow \text{i/p}$
 $\alpha \rightarrow \text{learning rate}$

$$w_{1, \text{new}} = w_{1, \text{old}} + \alpha * \text{error} * x_1$$

$$w_{2, \text{new}} = w_{2, \text{old}} + \alpha * \text{error} * x_2$$

$$w_{0, \text{new}} = w_{0, \text{old}} + \alpha * \text{error} * \text{bias}$$

first i/p instance

$$x_1 = 0; \quad x_2 = 0; \quad b = 1 \quad y = 0$$

$$w_1 = 0.2, \quad w_2 = 0, \quad w_0 = -0.5$$

x_1	x_2	Bias	y	y_{out}	Error	w_1 (0.2)	w_2 (0.3)	w_0 (-0.5)
0	0	1	0	0	0	0.2	0.3	-0.5
0	1	1	1	0	1	0.2	0.4	-0.1
1	0	1	1	1	0	0.2	0.4	-0.1
1	1	1	1	1	0	0.2	0.4	-0.1

$$y_{out} = \begin{cases} 1 & f(x) > 0 \\ 0 & f(x) \leq 0 \end{cases}$$

First i/p instance

$$f(x) = w_1 x_1 + w_2 x_2 + w_0$$

$$= 0 + 0 + (-0.5)$$

$$\boxed{f(x) = -0.5} \quad \boxed{y_{out} = 0} \rightarrow \text{Based on activation fn.}$$

$$\text{Error} = y - y_{out} = 0 - 0 = 0$$

$$\boxed{\text{Error} = 0} \quad \text{No need to update weight}$$

Update weights

$$w_1 = w_{old} + \alpha * \text{error} * x_1$$

$$w_{1, \text{new}} = 0.2 + 0.4(0) * 0 = 0.2$$

$$w_{2, \text{new}} = 0.3 + 0.4(0) * 1 = 0.3$$

$$b_{\text{new}} = -0.5 + 0.4(0) * 1 = -0.5$$

Second i/p instance

$$x_1 = 0 \quad x_2 = 1 \quad b = 1 \quad y = 1$$

$$w_1 = 0.2 \quad w_2 = 0.3$$

$$w_0 = -0.5$$

Summation

$$Y = 0.2(0) + 1(0.3) + (1)(-0.5)$$

$$\boxed{Y = -0.2}$$

$$\boxed{Y_{out} = 0}$$

$$\text{Error} = Y - Y_{out} = 1 - 0 = 1$$

$$\boxed{\text{Error} = 1}$$

$$W_{1\text{new}} = 0.2 + 0.4(1) * 0 = 0.2$$

$$W_{2\text{new}} = 0.3 + 0.4(1) * 1 = 0.7$$

$$W_{0\text{new}} = -0.5 + 0.4(1) * 1 = -0.1$$

Third input instance

$$x_1 = 1 \quad x_2 = 0 \quad b = 1 \quad y = 1 \quad w_1 = 0.2 \quad w_2 = 0.7$$

$$w_0 = -0.1$$

$$Y = 0.2(1) + 0.7(0) + (-0.1)(1)$$

$$\boxed{Y = 0.1}$$

$$\text{Activation } \boxed{Y_{out} = 1}$$

$$\text{error } Y - Y_{out} = 1 - 1 = 0$$

$$\boxed{\text{error} = 0}$$

No weight updation

Fourth Instance

$$x_1 = 1 \quad x_2 = 1 \quad b = 1 \quad y = 1 \quad w_1 = 0.2 \quad w_2 = 0.7$$

$$w_0 = -0.1 \quad \alpha = 0.4$$

$$Y = 0.2(1) + 0.7(1) - 0.1(1)$$

$$Y = 0.8$$

Activation

$$Y_{out} = 1$$

$$0.8 > 0$$

$$\text{Error (E)} = Y - Y_{out} = 1 - 1 = 0$$